Semi-Supervised Node Classification with Sparse Graph Attention

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Abstract

Graph neural networks have achieved strong results on node classification, yet they depend on large labelled sets. We propose a sparse attention mechanism that propagates label information more efficie Experiments on three citation benchmarks show consistent improvements over GCN and GAT baseline

Keywords: graph neural networks, attention, semi-supervised learning

1. Introduction

Learning on graph-structured data is central to many applications [1], [2]. Existing methods aggregate neighbour features uniformly, which we hypothesize limits performance on heterophilous graphs.

2. Method

We minimise a cross-entropy loss with an additional regularisation term. Training uses Adam (lr = 0.005

3. Results

Table 1 reports accuracy on Cora, Citeseer and Pubmed averaged over 10 runs ($p < 0.05$).

4. Conclusion

We introduced sparse graph attention and showed gains with fewer labels.

References

[1] T. Kipf and M. Welling, Semi-supervised classification with graph convolutional networks, ICLR, 201
[2] P. Velickovic et al., Graph attention networks, ICLR, 2018.